Unit IV: Rotational Motion and Gravitation

Torque, angular momentum, moment of inertia, gravitation, and motion of planets and satellites

Learning Goals

- Understand angular displacement, angular velocity, and angular acceleration.
- Relate linear motion quantities to rotational motion quantities.
- Define torque and use it to describe rotational effect of force.
- Understand moment of inertia as rotational inertia.
- Apply angular momentum and its conservation.
- Use Newton’s law of gravitation to calculate gravitational force.
- Understand gravitational field, gravitational potential energy, and orbital motion.
- Analyze basic motion of planets and satellites using gravitational force as centripetal force.

1. Introduction

Rotational motion is the motion of a body about an axis. A wheel, fan, spinning top, door, pulley, planet, and satellite all involve rotational ideas. Gravitation explains the attractive force between masses and is responsible for falling bodies, planetary motion, tides, and satellite orbits.

This unit connects two major ideas: how objects rotate and how gravity controls large-scale motion. Planetary and satellite motion can be understood by combining circular motion with Newton’s law of gravitation.

2. Angular Quantities

For rotational motion, we describe position using angular displacement instead of linear displacement. Angular displacement is the angle through which a body rotates. It is usually measured in radians.

```
theta = s / r        1 revolution = 2 pi radians
```

Linear quantity	Rotational quantity
Displacement s	Angular displacement theta.
Velocity v	Angular velocity omega.
Acceleration a	Angular acceleration alpha.
Mass m	Moment of inertia I.
Force F	Torque tau.
Momentum p	Angular momentum L.

Angular Velocity and Angular Acceleration

Angular velocity is the rate of change of angular displacement. Angular acceleration is the rate of change of angular velocity.

```
omega = Delta theta / Delta t        alpha = Delta omega / Delta t
```

- Angular velocity is measured in rad/s.
- Angular acceleration is measured in rad/s^2.
- If angular velocity is constant, angular acceleration is zero.

3. Relation Between Linear and Angular Motion

A point on a rotating body also has linear motion along a circular path. The farther the point is from the axis, the greater its linear speed for the same angular speed.

```
s = r theta        v = r omega        a_t = r alpha
```

```
Centripetal acceleration: a_c = v^2/r = r omega^2
```

- All points on a rigid rotating body have the same angular velocity.
- Points farther from the axis have larger linear speed.
- Tangential acceleration changes speed; centripetal acceleration changes direction.

4. Rotational Equations of Motion

When angular acceleration is constant, rotational motion equations resemble the equations of straight-line motion with linear quantities replaced by angular quantities.

```
omega = omega_0 + alpha t
```

```
theta = omega_0 t + (1/2) alpha t^2
```

```
omega^2 = omega_0^2 + 2 alpha theta
```

Linear equation	Rotational equation
v = u + at	omega = omega_0 + alpha t
s = ut + (1/2)at^2	theta = omega_0 t + (1/2)alpha t^2
v^2 = u^2 + 2as	omega^2 = omega_0^2 + 2 alpha theta

5. Torque

Torque is the turning effect of a force. A force can produce rotation more effectively when it is applied farther from the axis or at a better angle. Opening a door is easier when force is applied at the handle rather than near the hinge.

```
Torque: tau = r F sin(theta)
```

Symbol	Meaning
tau	Torque, measured in newton metre (N m).
r	Perpendicular distance or lever arm from axis to line of action of force.
F	Applied force.
theta	Angle between r and F.

- Torque is maximum when force is perpendicular to the lever arm.
- Torque is zero when the line of action of force passes through the axis.
- Clockwise and anticlockwise torques are usually assigned opposite signs.

Rotational Form of Newton’s Second Law

For rotation about a fixed axis, net torque is equal to moment of inertia times angular acceleration.

```
tau_net = I alpha
```

- Greater torque produces greater angular acceleration for the same moment of inertia.
- Greater moment of inertia makes rotational acceleration harder for the same torque.

6. Moment of Inertia

Moment of inertia is the rotational analogue of mass. It measures how difficult it is to change an object's rotational motion. It depends not only on mass but also on how the mass is distributed relative to the axis of rotation.

```
For point masses: I = sum m_i r_i^2
```

Object	Moment of inertia about common axis
Point mass at distance r	I = mr^2
Thin ring about central axis	I = MR^2
Solid disc about central axis	I = (1/2)MR^2
Solid sphere about diameter	I = (2/5)MR^2
Thin rod about center	I = (1/12)ML^2

- Mass farther from the axis contributes more strongly because of the r^2 factor.
- A ring has greater moment of inertia than a solid disc of the same mass and radius.
- Figure skaters spin faster when they pull their arms inward because moment of inertia decreases.

7. Rotational Kinetic Energy

A rotating body has kinetic energy due to rotation. The formula resembles translational kinetic energy, with mass replaced by moment of inertia and linear speed replaced by angular speed.

```
Rotational kinetic energy: K_rot = (1/2) I omega^2
```

```
Total kinetic energy for rolling: K = (1/2)Mv^2 + (1/2)I omega^2
```

- Pure rotation has rotational kinetic energy.

- Rolling motion has both translational and rotational kinetic energy.
- For rolling without slipping, v = R omega.

8. Angular Momentum

Angular momentum describes the rotational motion of a body. For a particle, it depends on position and linear momentum. For a rigid body rotating about a fixed axis, it depends on moment of inertia and angular velocity.

```
For rigid body: L = I omega
```

```
For particle: L = r x p
```

- Angular momentum is a vector quantity.
- The direction of angular momentum is given by the right-hand rule.
- Torque changes angular momentum.

```
tau_net = dL/dt
```

Conservation of Angular Momentum

If net external torque on a system is zero, total angular momentum remains constant. This principle explains many natural and technological phenomena.

```
If tau_net = 0, then L = constant
```

- A spinning skater rotates faster when pulling arms inward.
- Planets move faster when closer to the Sun and slower when farther away.
- Rotating satellites and gyroscopes use angular momentum for stability.

9. Equilibrium of Rigid Bodies

For a rigid body to be in complete equilibrium, both translational and rotational conditions must be satisfied. The net force must be zero and the net torque must be zero.

```
Translational equilibrium: sum F = 0        Rotational equilibrium: sum tau = 0
```

- A balanced seesaw has clockwise torque equal to anticlockwise torque.
- A ladder leaning against a wall is analyzed using force and torque balance.
- The choice of axis for torque calculation can simplify the problem.

10. Newton’s Law of Gravitation

Newton’s universal law of gravitation states that every mass attracts every other mass with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers.

```
F = G m1 m2 / r^2
```

Symbol	Meaning
F	Gravitational force.
G	Universal gravitational constant.
m1, m2	Masses of the two bodies.
r	Distance between centers of the bodies.

- Gravity is always attractive.
- The force acts along the line joining the centers of the two masses.
- If distance doubles, gravitational force becomes one-fourth.
- The forces on the two bodies are equal in magnitude and opposite in direction.

11. Gravitational Field and Acceleration Due to Gravity

The gravitational field at a point is the gravitational force per unit mass experienced by a small test mass placed at that point.

```
g = F/m = GM/r^2
```

Near Earth’s surface, g is approximately 9.8 m/s^2. At height h above Earth’s surface, the distance from Earth’s center is R + h, so g decreases with altitude.

```
g_h = GM / (R + h)^2
```

- g is larger near massive bodies.
- g decreases as distance from the center of the planet increases.
- Weight is the gravitational force on a body: W = mg.

12. Gravitational Potential Energy

For two masses separated by distance r, gravitational potential energy is negative if zero is chosen at infinite separation. The negative sign shows that gravity is attractive and energy must be supplied to separate the masses to infinity.

```
U = -G M m / r
```

Near Earth’s surface, when height changes are small compared with Earth’s radius, the simpler formula U = mgh is a good approximation.

```
Near Earth’s surface: Delta U = mg Delta h
```

13. Motion of Planets and Satellites

A satellite remains in orbit because gravity provides the centripetal force needed for circular or nearly circular motion. The satellite is continuously falling toward Earth while also moving forward fast enough to keep missing Earth.

```
Gravitational force = centripetal force
```

```
GMm/r^2 = mv^2/r
```

```
Orbital speed: v = sqrt(GM/r)
```

- Orbital speed does not depend on the mass of the satellite.
- A satellite closer to Earth needs greater orbital speed.
- A higher orbit has lower speed but longer period.

Time Period of a Circular Orbit

```
T = 2 pi r / v        T^2 = 4 pi^2 r^3 / GM
```

This relation shows that the square of the orbital period is proportional to the cube of orbital radius. It is connected to Kepler’s third law of planetary motion.

14. Kepler’s Laws

Kepler’s laws describe planetary motion around the Sun. Newton’s law of gravitation later explained why these laws work.

- First law: Planets move in elliptical orbits with the Sun at one focus.
- Second law: A line joining a planet and the Sun sweeps out equal areas in equal times.
- Third law: The square of the orbital period is proportional to the cube of the semi-major axis.

```
Kepler’s third law: T^2 proportional to r^3 for circular orbit approximation
```

Kepler’s second law is a consequence of conservation of angular momentum. When a planet is closer to the Sun, it moves faster; when farther away, it moves slower.

15. Worked Examples

Example 1: Torque

A force of 20 N is applied perpendicular to a door at a distance 0.5 m from the hinge. Torque tau = rF sin(90 degrees) = 0.5 x 20 = 10 N m.

Example 2: Moment of Inertia

A point mass of 2 kg is located 3 m from the axis. Moment of inertia I = mr^2 = 2 x 3^2 = 18 kg m^2.

Example 3: Angular Momentum

A wheel has moment of inertia 4 kg m^2 and angular speed 5 rad/s. Angular momentum L = I omega = 4 x 5 = 20 kg m^2/s.

Example 4: Gravitational Force

For two masses m1 and m2 separated by distance r, gravitational force is F = Gm1m2/r^2. If r becomes twice as large, the new force becomes one-fourth of the original force.

Example 5: Orbital Speed

For a satellite in circular orbit of radius r around Earth of mass M, set gravity equal to centripetal force: GMm/r^2 = mv^2/r. Solving gives v = sqrt(GM/r).

16. Common Mistakes

- Confusing angular velocity with linear velocity.
- Forgetting that radians are dimensionless but important in formulas.
- Calculating torque using full distance instead of perpendicular lever arm.
- Thinking moment of inertia depends only on mass; it also depends on mass distribution.
- Assuming angular momentum is always conserved even when external torque acts.
- Calling centripetal force a new type of force instead of the net inward force.
- Using U = mgh for very large height changes where the inverse-square nature of gravity matters.
- Forgetting that satellite orbital speed is determined by orbital radius and central mass, not satellite mass.

Quick Revision Sheet

- Angular displacement is measured in radians.
- s = r theta, v = r omega, and a_t = r alpha.
- Torque is tau = rF sin(theta).
- Net torque produces angular acceleration: tau_net = I alpha.
- Moment of inertia measures resistance to rotational acceleration.
- Rotational kinetic energy is (1/2)I omega^2.
- Angular momentum for a rigid body is L = I omega.
- Angular momentum is conserved when net external torque is zero.
- Newton’s law of gravitation is F = Gm1m2/r^2.
- Gravitational field is g = GM/r^2.
- Gravitational potential energy is U = -GMm/r.
- For circular orbit, gravity provides centripetal force.
- Orbital speed is v = sqrt(GM/r).
- Kepler’s second law follows from conservation of angular momentum.

Practice Questions

- Define angular displacement, angular velocity, and angular acceleration.
- Convert one complete revolution into radians.
- A wheel of radius 0.4 m rotates with angular speed 10 rad/s. Find linear speed at the rim.
- A force of 30 N acts perpendicular to a lever arm of 0.2 m. Find torque.
- Why is it easier to open a door by pushing at the handle rather than near the hinge?

- Define moment of inertia. Why does mass distribution matter?
- Find the moment of inertia of a 3 kg point mass at a distance 2 m from the axis.
- State the law of conservation of angular momentum with an example.
- What are the two conditions for equilibrium of a rigid body?
- State Newton’s universal law of gravitation.
- If the distance between two masses is tripled, how does gravitational force change?
- Define gravitational field intensity.
- Why is gravitational potential energy negative when zero is taken at infinity?
- Derive the expression v = sqrt(GM/r) for orbital speed.
- State Kepler’s three laws of planetary motion.
- Why does a planet move faster when it is closer to the Sun?